

Lesson 5: Raster Analysis

Practical Session: Multiband Rasters, RGB Composites, and Band Ratios in QGIS

Objectives

1. Understand the difference between **single-band** and **multiband** rasters, and how QGIS maps individual bands onto the Red, Green, and Blue display channels.
2. Manipulate the band-to-channel assignment of the Natural Earth hypsometric raster to produce True Colour and False Colour composites.
3. Load and explore a real multiband **Landsat 9** scene and build RGB composites from its spectral bands using a **virtual raster**.
4. Apply the concept of a **band ratio** by calculating the **NDVI** (Normalized Difference Vegetation Index) with the Raster Calculator.

Preliminary Setup

1. Launch QGIS and initialise a new, empty project.
2. Confirm the Project CRS in the bottom-right status bar reads EPSG:4326 (WGS 84).
3. From your Lesson_1/Data folder, drag in the **HYP_50M_SR_W.tif** raster (Cross Blended Hypso with Shaded Relief and Water). You worked with this file in Lessons 2 and 4.
4. Create a new, empty folder named Lesson_5/Data on your computer. The Landsat scene you download and every new file you create today will be saved here.

Data

The Landsat dataset for this session has already been prepared for you. **Do not download anything from the [USGS EarthExplorer](#) portal today** — it would consume the entire session. Instead, download the pre-packaged scene from the **FU-Box** [link](#) provided to you, unzip it, and place the folder inside your Lesson_5/Data directory. The archive contains the individual spectral bands of a single **Landsat 8/9 OLI** scene as separate GeoTIFF files (one file per band, named ..._B2.TIF, ..._B3.TIF, ..._B4.TIF, ..._B5.TIF, and so on).

A Very Short Introduction to Landsat: The Landsat programme is a joint NASA/USGS mission that has continuously imaged Earth's surface since 1972 — the longest-running record of the planet from space. The current satellites, Landsat 8 (2013) and Landsat 9 (2021), carry the OLI sensor, which records reflected sunlight in several separate **spectral bands** — narrow slices of the electromagnetic spectrum. Crucially, some of these bands lie outside the range of human vision, such as the Near-Infrared (NIR) band. Because vegetation reflects NIR very strongly, these invisible bands carry information we cannot see with our eyes but can reveal by feeding them into the display channels ourselves.

Step 1: Single-band vs. Multiband Rasters

A raster is simply a grid of pixels, and each pixel holds one or more **values**. A **single-band** raster stores exactly one value per pixel — for example an elevation model, where each pixel is a height in metres. A **multiband** raster stores several values per pixel, stacked like transparent sheets. The Natural Earth `HYP_50M_SR_W.tif` is a multiband raster: it holds three bands (Red, Green, Blue) which together produce the familiar coloured relief image you see on screen.

Procedure

1. Right-click `HYP_50M_SR_W` in the Layers Panel and choose **Properties → Symbolology**.
2. At the top, note that the **Render type** is set to **Multiband color**. Below it you will see three dropdowns: **Red band**, **Green band**, and **Blue band**. By default these are assigned to **Band 1**, **Band 2**, and **Band 3** respectively.
3. This default assignment — putting each real colour band into its matching display channel — is called a **True Colour Composite (TCC)**. The Earth looks "natural": blue oceans, green vegetation, brown mountains.

Key Concept — Bands vs. Channels: A raster stores **bands** (the recorded data). Your screen only has three **channels** it can display: Red, Green, and Blue. RGB visualization is the act of deciding which band goes into which channel. The data never changes — only the instructions we give the screen do.

Step 2: Swapping Bands — True Colour vs. False Colour

Because RGB visualization is just a set of instructions, we are free to feed any band into any channel. Doing so changes how the planet *looks* without altering a single pixel of the underlying data.

Procedure

1. Staying in **Properties → Symbolology** for `HYP_50M_SR_W`, change the band assignments as follows:
 - a. **Red band** → Band 3
 - b. **Green band** → Band 1
 - c. **Blue band** → Band 2
2. Click **Apply**. Observe how the continents shift into surreal, unnatural colours. This is a **False Colour Composite (FCC)** — the same data, displayed through a different set of instructions.
3. Experiment freely: try a few more combinations (e.g. Red → Band 2, Green → Band 3, Blue → Band 1) and click **Apply** after each. Watch the Earth change colour each time.
4. When you are finished exploring, return the assignments to their defaults (**Red → Band 1**, **Green → Band 2**, **Blue → Band 3**) and click **OK** to restore the True Colour Composite.

Why This Matters: On the Natural Earth raster this is only a fun visual trick, because all three bands are ordinary visible colours. But on a real satellite image like Landsat — where one of the bands is invisible Near-Infrared — swapping bands into the display channels reveals genuinely new information, as you will see in Step 4.

Step 3: Loading the Landsat Scene

The Landsat scene you downloaded from FU-Box arrives as a set of **separate single-band files** — one GeoTIFF per spectral band. Each file on its own is a grey-scale single-band raster. To display them in colour, we first load the ones we need and then combine them.

Procedure

1. Go to **Layer → Add Layer → Add Raster Layer...** and browse to your Landsat folder inside Lesson_5/Data.
2. Select the following four band files together (hold **Ctrl** to multi-select) and click **Add**:
 - a. **B2** — Blue
 - b. **B3** — Green
 - c. **B4** — Red
 - d. **B5** — Near-Infrared (NIR)
3. Each band loads as a separate grey-scale layer. Click through them in the Layers Panel and notice that individually they are just shades of grey — a single-band raster has no inherent colour.

The Landsat Band Cheat-Sheet (OLI sensor): Band 2 = Blue · Band 3 = Green · Band 4 = Red · Band 5 = Near-Infrared. Keeping this order in mind is the key to building the composites in the next step.

Step 4: Building RGB Composites with a Virtual Raster

To turn separate grey-scale bands into a single colour image, we stack them into one multiband layer. Rather than writing a large new file to disk, we use a **virtual raster (VRT)** — a tiny text file that simply tells QGIS "treat these existing bands as one stacked raster." It behaves exactly like a multiband file but takes up almost no space.

Procedure — True Colour Composite (TCC)

1. Open the **Processing Toolbox (Processing → Toolbox)**.
2. Search for **Build virtual raster** (a GDAL tool) and double-click to open it.
3. Click the ... button next to **Input layers** and tick bands **B4**, **B3**, and **B2**. Click OK.

4. Very important: check the box **Place each input file into a separate band**. Without this, the tool will mosaic the bands side-by-side instead of stacking them.
5. Under **Virtual**, save the output as `landsat_TCC.vrt` inside Lesson_5/Data, then click **Run**.
6. The new layer will still look odd at first (often very dark). Right-click it → **Properties** → **Symbology**, set **Red** → **Band 3**, **Green** → **Band 2**, **Blue** → **Band 1**, and under **Contrast enhancement** choose **Stretch to MinMax** and under **Min/Max Value Settings** choose **Min/Max**. Click OK. You now see the scene in natural True Colour.

Procedure — False Colour Composite (FCC)

1. Run **Build virtual raster** again, this time selecting bands **B5**, **B4**, and **B3** (again ticking **Place each input file into a separate band**).
2. Save it as `landsat_FCC.vrt` in Lesson_5/Data and click **Run**.
3. Apply the same enhancement settings in Symbology. In this composite, healthy vegetation glows in bright **red**, because we have fed the invisible Near-Infrared band (B5) into the Red display channel — and vegetation reflects NIR very strongly.

Standard FCC Recipe: The classic "colour-infrared" composite is NIR–Red–Green → R–G–B, i.e. Landsat bands **B5–B4–B3**. Whenever you see a satellite image where forests appear vivid red, this is the recipe being used.

Step 5: Band Ratios and NDVI

Swapping bands into display channels is powerful, but we can go further and do **arithmetic** on the band values themselves. A **band ratio** combines two bands into a single new value per pixel that isolates a specific surface property. The most famous is the **NDVI (Normalized Difference Vegetation Index)**, which measures how healthy and dense vegetation is by exploiting the fact that plants reflect a lot of NIR but absorb Red light.

The NDVI is defined as:

$$\text{NDVI} = (\text{NIR} - \text{Red}) / (\text{NIR} + \text{Red}) = (\text{Band 5} - \text{Band 4}) / (\text{Band 5} + \text{Band 4})$$

The result always falls between –1 and +1. Dense, healthy vegetation gives high positive values (roughly 0.6–0.9); bare soil, rock, and urban areas give low values near zero; and water is typically negative.

Procedure

1. Go to **Raster** → **Raster Calculator**....
2. In the **Raster Bands** list on the left you will see your loaded Landsat bands, each labelled with an @1 suffix (e.g. B5@1 means "band 1 of the B5 file").

3. Build the following expression by double-clicking band names and typing the operators. Substitute your actual file names for B5 and B4:

```
("B5@1" - "B4@1") / ("B5@1" + "B4@1")
```

4. Set the **Output layer** to landsat_NDVI.tif inside Lesson_5/Data.
5. Leave the output extent and resolution at their defaults (they will match your input bands), then click **OK**. The NDVI raster is calculated and added to the map — at first it appears as a grey-scale single-band image.
6. To make the vegetation readable, right-click the NDVI layer → **Properties** → **Symbology**. Change **Render type** to **Singleband pseudocolor**, pick a **green** colour ramp (e.g. "Greens", or "RdYlGn" for red-to-green), and click **Classify**, then **OK**. Bright green areas are now healthy vegetation; pale or red areas are bare ground and water.

Note on Single-band Output: Notice that NDVI is a *single-band* raster — one calculated value per pixel — even though it was built from two bands. This brings us full circle to Step 1: we began by taking multiband data apart into channels, and we end by combining bands arithmetically into a brand-new single-band result.

Summary

1. **Bands Are Data, Channels Are Display** — A single-band raster holds one value per pixel; a multiband raster holds several. RGB visualization is simply the choice of which band feeds each of the screen's three colour channels, and changing it never alters the underlying data.
2. **True vs. False Colour** — A True Colour Composite places the natural R, G, B bands in their matching channels. A False Colour Composite deliberately re-routes bands — most usefully the invisible Near-Infrared — to reveal information the eye cannot see, such as vegetation glowing red.
3. **Virtual Rasters Stack Bands Cheaply** — Build virtual raster combines separate single-band files into one multiband layer without duplicating the data on disk, which is how we assembled the Landsat composites.
4. **Band Ratios Extract Meaning** — NDVI, computed with the Raster Calculator as $(\text{NIR} - \text{Red}) / (\text{NIR} + \text{Red})$, turns two bands into a single index of vegetation health — the foundation of quantitative remote sensing.

Homework: Mapping Vegetation Health

Objective

Produce a two-map layout that contrasts what the human eye sees with what the vegetation index reveals, reinforcing the difference between a colour composite and a band ratio.

Procedure

1. Ensure your True Colour Composite (`landsat_TCC.vrt`) and your `landsat_NDVI.tif` layer are both loaded and styled as in Steps 4 and 5 (the NDVI in a green pseudocolor ramp).
2. Navigate to **Project → New Print Layout** and name it **Homework_5**.
3. Set the page orientation to **Landscape**.
4. Use the **Add Map** tool to draw a map frame on the **left** half of the page showing the True Colour Composite.
5. Use **Add Map** again to draw a second frame on the **right** half showing the NDVI layer, at the same scale and extent.
6. Add a title, a legend (for the NDVI colour ramp), a scale bar, and a north arrow.
7. Export the final layout as a PDF via **Layout → Export as PDF....**

Submission Deliverable

Submit the exported single-page PDF layout showing the True Colour and NDVI maps side by side.